



Computing of Society: Why It Matters and How It Can Be Done

*Saarbrücken Summer Institute in Computational Social Science
(SICSS), 15.09.2025*

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de Madrid

Before we start: ...your background?

- Social or Computer Scientist?
- Student, PhD researcher, postdoctoral researcher?





Computing of Society

- 1) Why It Matters
- 2) How It Can Be Done





Computing of Society

1) Why It Matters

2) How It Can Be Done

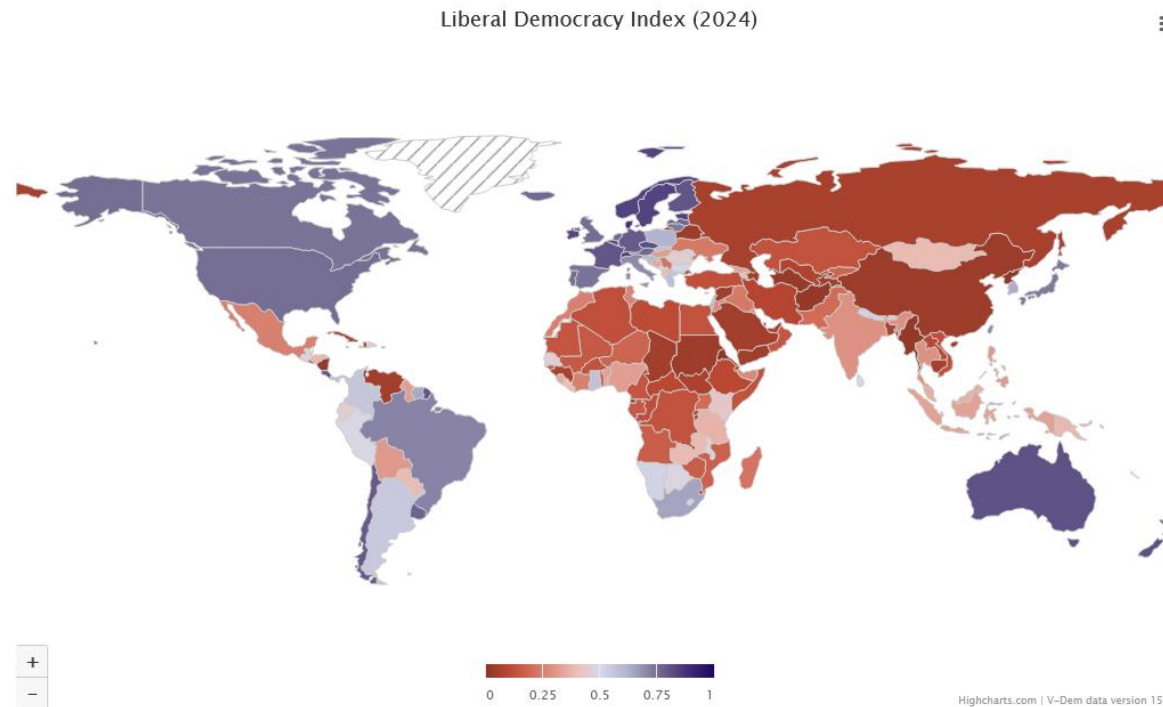


Computing of Society: Why does it matter?

- As political scientists, we are naturally interested in social and political processes.
- In political science, we describe political phenomena and investigate their causes and consequences.

Computing of Society: Why does it matter?

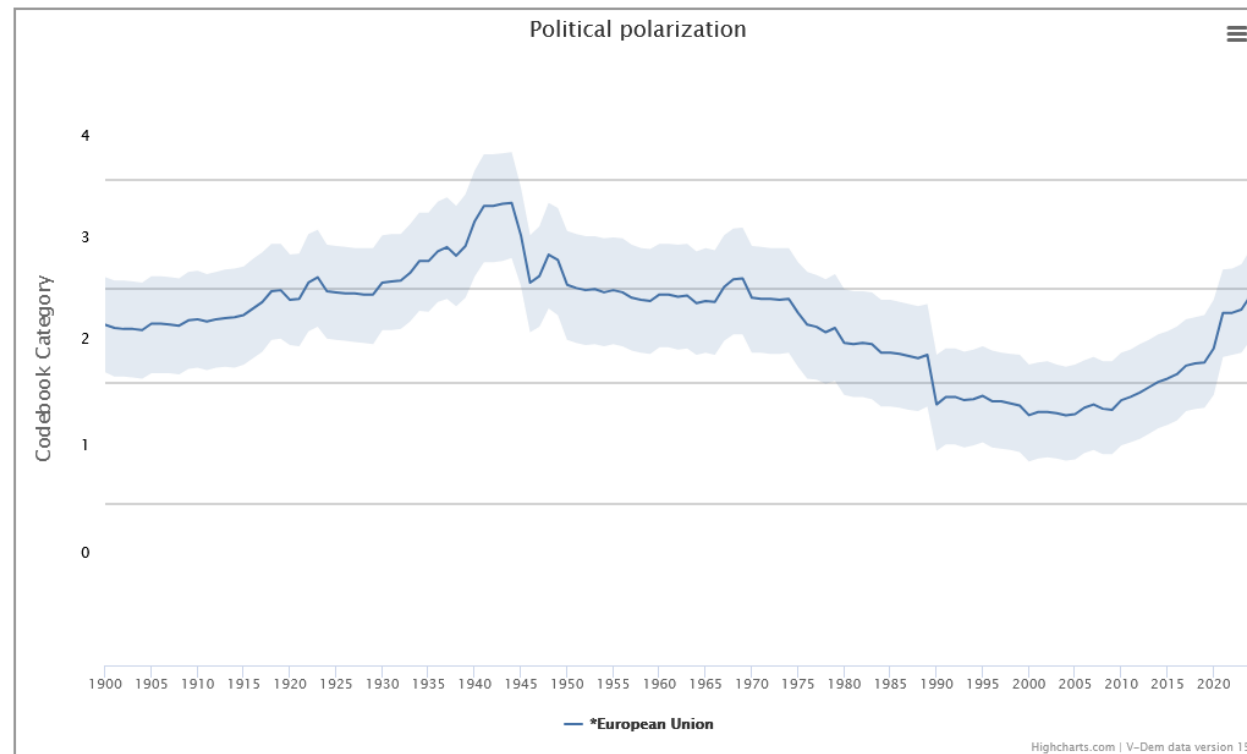
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- Examples



Source: https://v-dem.net/data_analysis/VariableGraph/ (27.05.2025).

Computing of Society: Why does it matter?

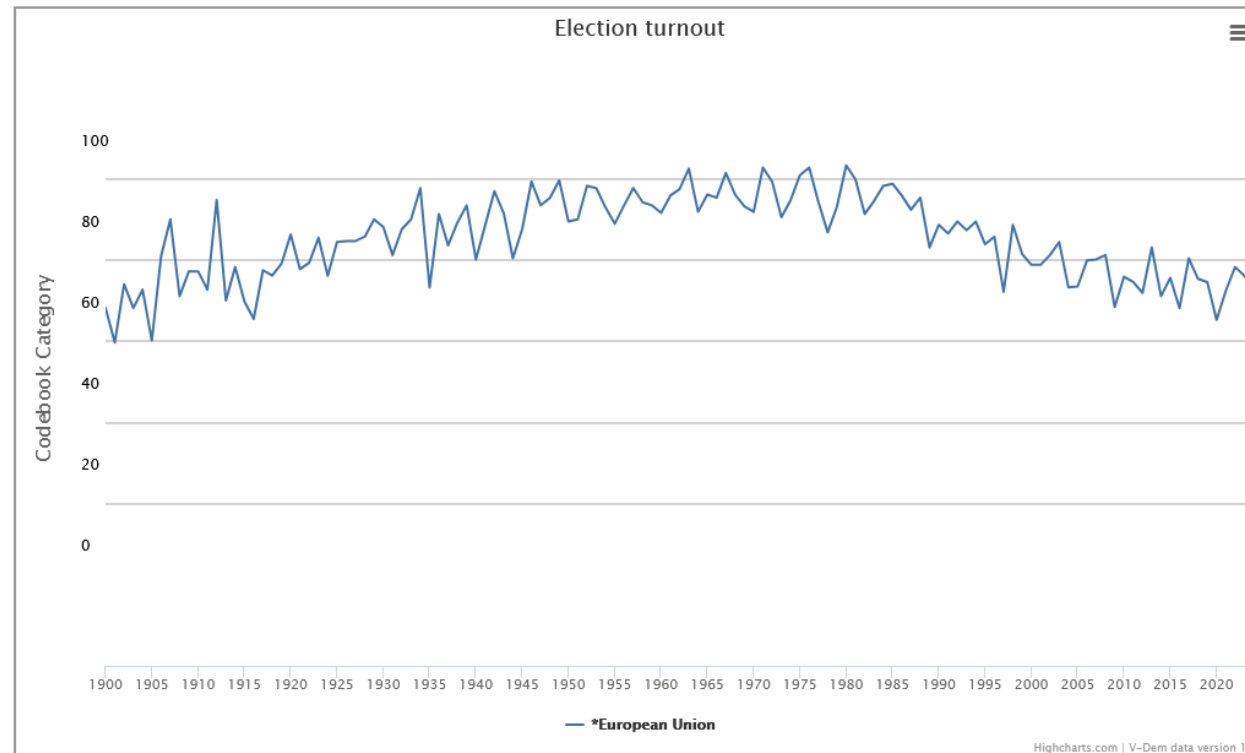
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Computing of Society: Why does it matter?

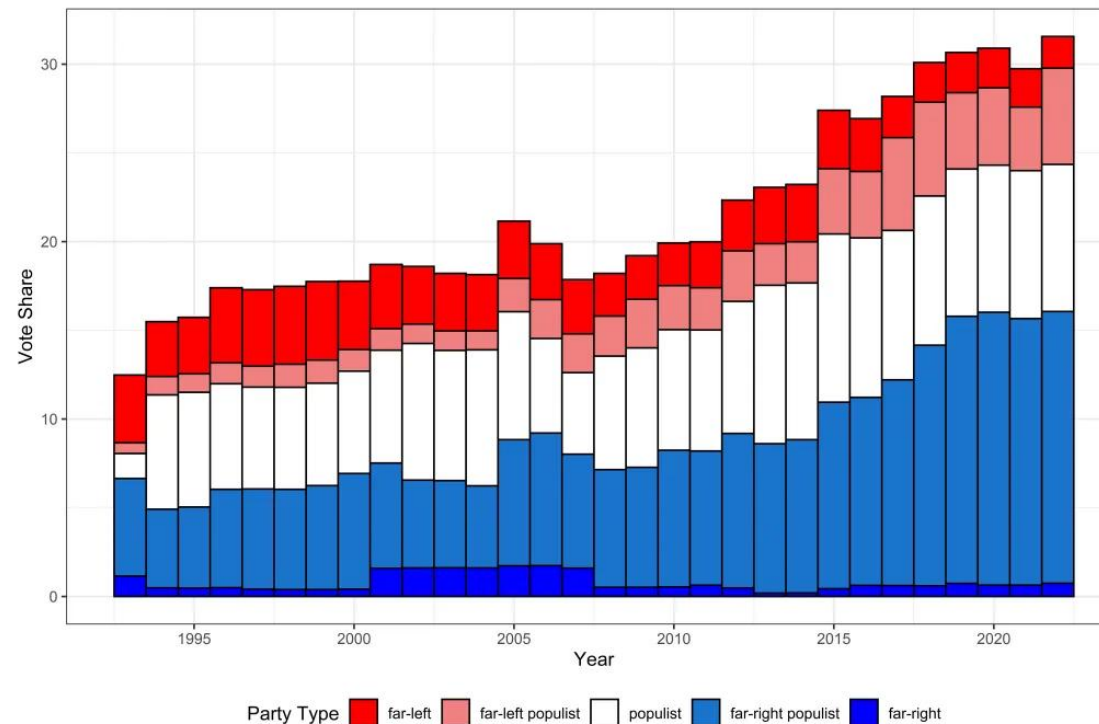
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Computing of Society: Why does it matter?

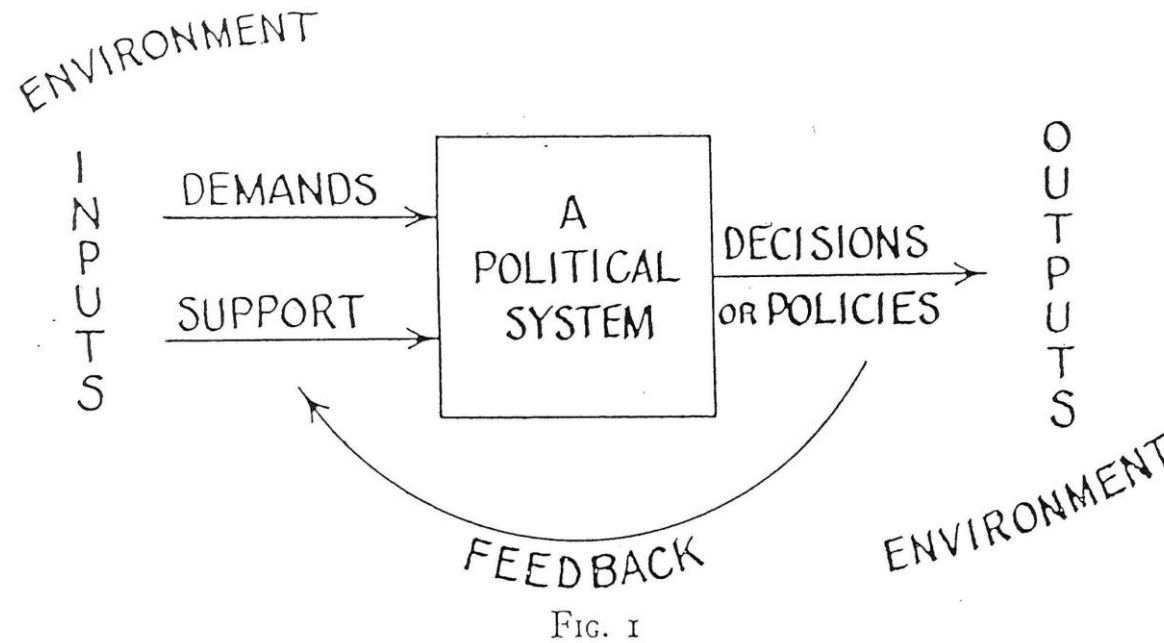
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Source: The PopuList data set: <https://doi.org/10.17605/OSF.IO/2EWKQ>.

Computing of Society: Why does it matter?

- As political scientists, we are naturally interested in social and political processes.
- In political science, we describe political phenomena and investigate their causes and consequences – **based on theory/ theoretical assumptions.**



Quelle: Easton, David (1957): An Approach to the Analysis of Political Systems. World Politics, 9, 383-400.



Computing of Society

1) Why It Matters

2) How It Can Be Done

- Traditional data sources/methods
- Innovative/digital data sources/methods





Computing of Society

1) Why It Matters

2) How It Can Be Done

- Traditional data sources/methods
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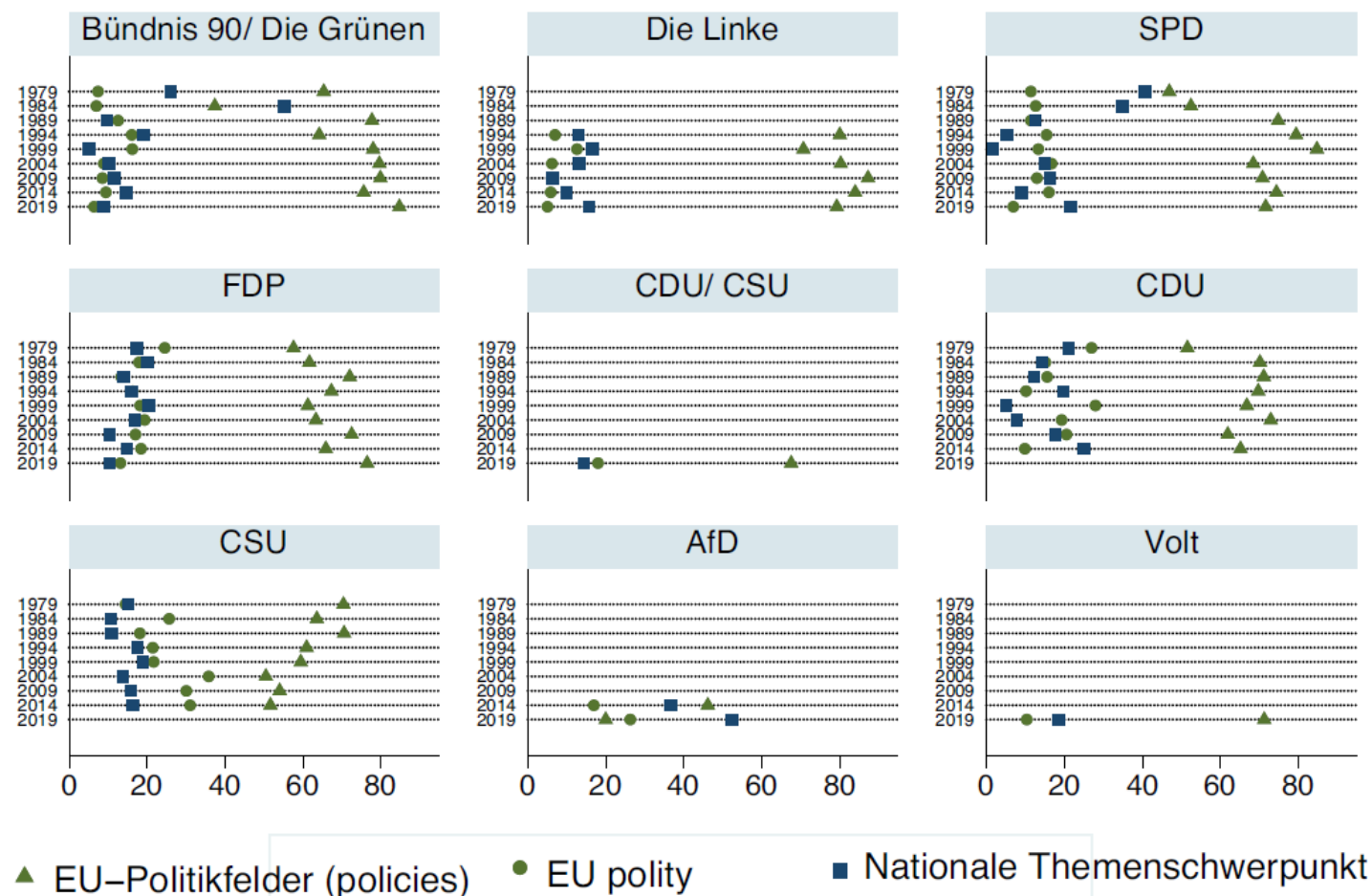


How It Can Be Done: traditional ways

Research topic:
Party competition
in Europe

Data:
Party manifestos

Research question:
How salient are
European issues for
political parties in
EP elections?



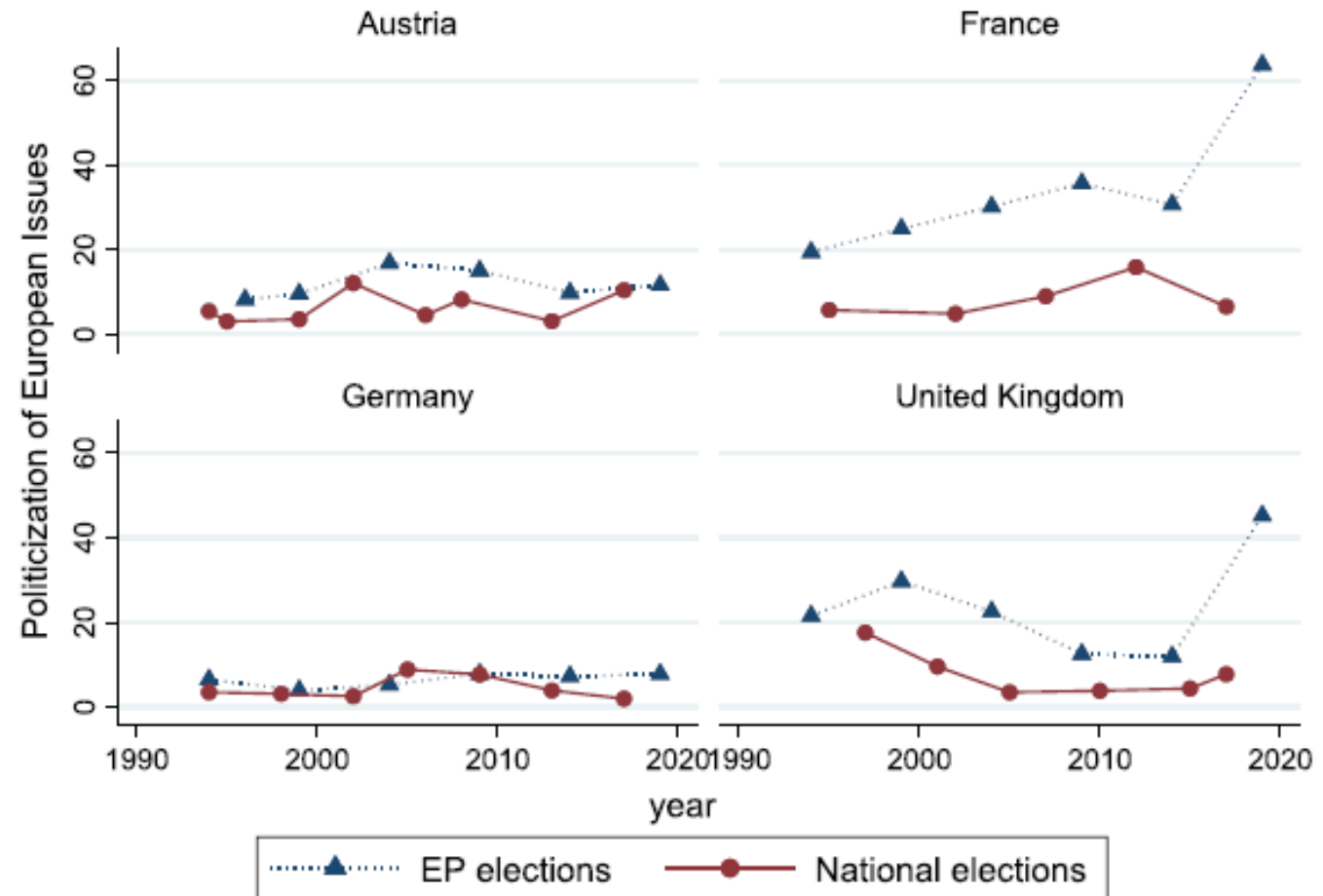
Source: Braun, Daniela and Markus Tausendpfund (2023): Wie europäisch sind die Europawahlen? In: Deutschland & Europa: Europa im Wandel – EU-Politik und Fußball-EM im Rampenlicht Heft 86: 8-14.

How It Can Be Done: traditional ways

Research topic:
Election campaigns
in Europe

Data:
Newspaper articles

Research question:
How important are
European in
national and EP
elections?



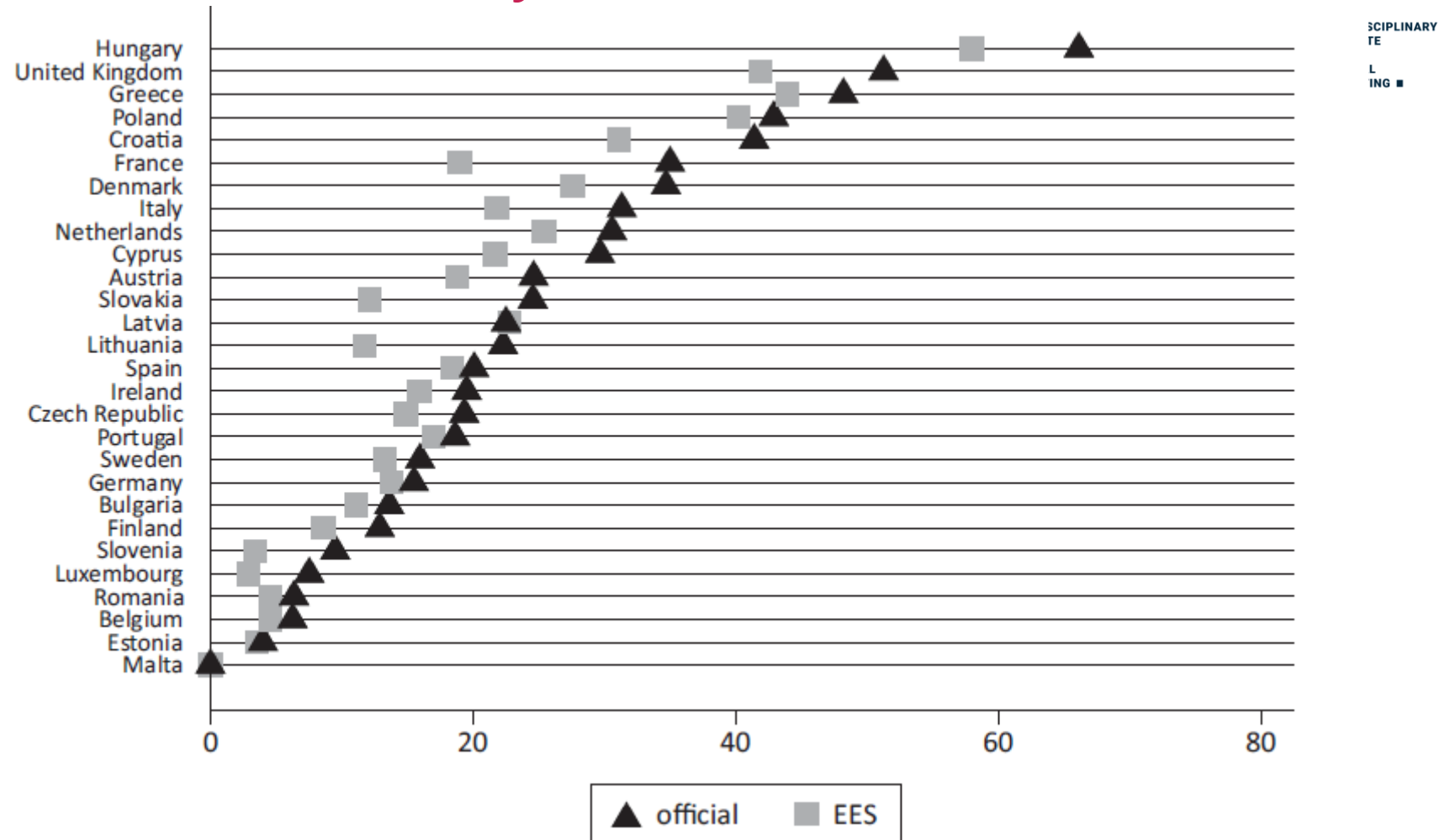
Source: Braun, Daniela und Edgar Grande (2021): The politicization of elections to the European Parliament. The crucial role of mainstream political parties. *Journal of Common Market Studies* 59(5): 1124-1141.

How It Can Be Done: traditional ways

Research topic:
Political participation

Data: Public opinion
surveys

Research question:
Who participates
why in EP elections?



Source: Braun, Daniela and Markus Tausendpfund (2020): Electoral behaviour in a European Union under stress. In: Politics and Governance 8 (1): 28-40.

How It Can Be Done: traditional ways

Research topics:

Political participation
and attitudes

Data: Public opinion
surveys

[ESS; EB; EES; ISSP;
EVS/WVS and many
others]

How It Can Be Done: traditional ways

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The ActEU public opinion survey

- fieldwork: May-June 2024
- data for 10 EU member states
- questions on political attitudes and political behaviour
- survey innovation on political trust
- survey innovation on affective issue polarization
- experimental elements of the survey
- Access via: GESIS Leibniz Institute for the Social Sciences, <https://doi.org/10.7802/2782>

ActEU Activating European
Citizens' Trust in Times of Crises and Polarisation



Funded by
the European Union



Computing of Society

1) Why It Matters

2) How It Can Be Done

- Traditional data sources/methods
- Innovative/digital data sources/methods



How It Can Be Done: more innovative ways

Why are innovative data and methods needed?

- Traditional data sources/methods are expensive and time-consuming.
 - Data collection takes time and requires a lot of human resources (e.g. human coding approach in manifesto research)
 - Data cleaning and reliability checks as an additional necessary step
 - Only after this, data can be used to answer research questions
- Scientific findings are never available in real time, but always with enormous time delays, and are then often not free of errors

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- Traditional data sources only map the offline world – but what about the online world where citizens, voters and political actors are interacting too?

How It Can Be Done: more innovative ways

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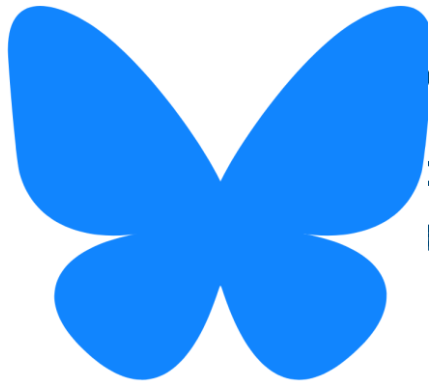
- Traditional data sources/methods are expensive and time-consuming.

- Data collection approach in human resources

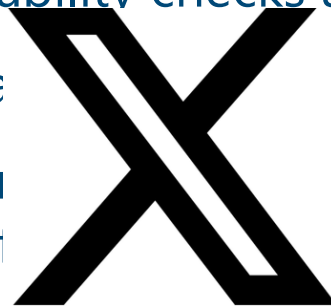
facebook

TikTok

- Data cleaning and reliability checks as an additional necessary step



After this, data can be analyzed and findings are identified and are then often used for research.



Answer research questions in real time and avoid errors



with enormous data

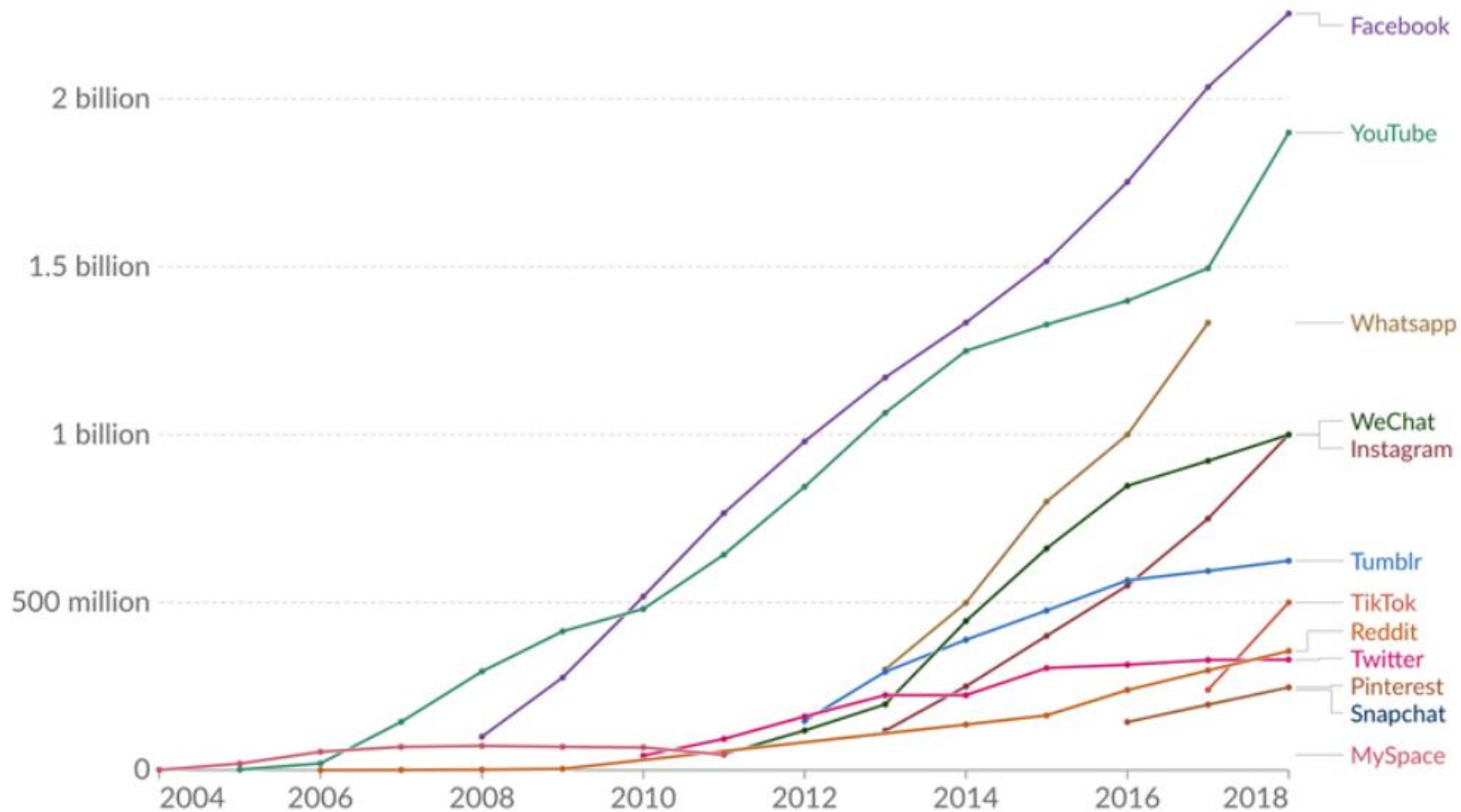


- Traditional data sources only map the offline world – but what about the online world where citizens, voters and political actors are interacting too?

Number of people using social media platforms, 2004 to 2018

Estimates correspond to monthly active users (MAUs). Facebook, for example, measures MAUs as users that have logged in during the past 30 days. See source for more details.

Our World
in Data



Source: Statista and TNW (2019)

OurWorldInData.org/internet • CC BY

Beyond innovative data, politics have changed



Elon Musk  
@elonmusk

You are the media now

[Traducir post](#)

7:17 a. m. · 6 nov. 2024 · **89,1 M** Reproducciones

 43 mil

 142 mil

 1 M

Beyond innovative data, politics have changed



Elon Musk
@elonmusk

You are the media now

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From online influencer to MEP: Meet the EU's most connected lawmakers



Copyright ASSOCIATED PRESS and Twitter (X)

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43 mil

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1 M



Donald J. Trump
@realDonaldTrump

[Follow](#)

I am officially running for President of the United States. [#MakeAmericaGreatAgain](#)
[instagram.com/p/3_vGGrmhdq/](https://www.instagram.com/p/3_vGGrmhdq/)

Retweets
11,986

Likes
12,378



8:57 AM · 16 Jun 2015

2.2K

12K

12K

From online influencer to MEP: Meet the EU's most connected lawmakers



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Article | [Open access](#) | Published: 09 January 2023

Exposure to the Russian Internet Research Agency foreign influence campaign on Twitter in the 2016 US election and its relationship to attitudes and voting behavior

[Gregory Eady](#) , [Tom Paskhalis](#) , [Jan Zilinsky](#), [Richard Bonneau](#), [Jonathan Nagler](#) & [Joshua A. Tucker](#)

[Nature Communications](#) **14**, Article number: 62 (2023) | [Cite this article](#)

70k Accesses | 44 Citations | 2021 Altmetric | [Metrics](#)

Abstract

There is widespread concern that foreign actors are using social media to interfere in elections worldwide. Yet data have been unavailable to investigate links between exposure to foreign influence campaigns and political behavior. Using longitudinal survey data from US respondents linked to their Twitter feeds, we quantify the relationship between exposure to the Russian foreign influence campaign and attitudes and voting behavior in the 2016 US election. We demonstrate, first, that exposure to Russian disinformation accounts was heavily concentrated: only 1% of users accounted for 70% of exposures. Second, exposure was concentrated among users who strongly identified as Republicans. Third, exposure to the Russian influence campaign was eclipsed by content from domestic news media and politicians. Finally, we find no evidence of a meaningful relationship between exposure to the Russian foreign influence campaign and changes in attitudes, polarization, or voting behavior. The results have implications for understanding the limits of election interference campaigns on social media.

The New York Times

[Russia-Ukraine War](#) > [The Latest](#) [Photos](#) [Maps](#) [Policing During Wartime](#) [What is the Oreshnik Ballistic Missile?](#) [Putin's Chilling Message](#)

Russia Takes Its Ukraine Information War Into Video Games

Propaganda is appearing in Minecraft and other popular games and discussion groups as the Kremlin tries to win over new audiences.

 Share full article  



In the Russian version of the World of Tanks video game, a recreation of the Soviet Union's parade of tanks in Moscow in 1945. [via YouTube](#)



By [Steven Lee Myers](#) and [Kellen Browning](#)

Reporting from San Francisco

July 30, 2023

Digital technologies has been linked to political polarisation because of their ability to foster echo chambers that amplify extremist ideas (Sunstein 2018)

Why was social media praised as the great equalizer?

Digital technologies has been linked to political polarisation because of their ability to foster echo chambers that amplify extremist ideas (Sunstein 2018)

Why was social media praised as the great equalizer?



At 22, I was working w/ children + communities.

At 25, I was waitressing to support my family.

At 28, I won my primary for Congress. I didn't have health insurance.

Today I turn 29.

You're never too late, too early, or too imperfect to care for yourself or pursue your dreams.

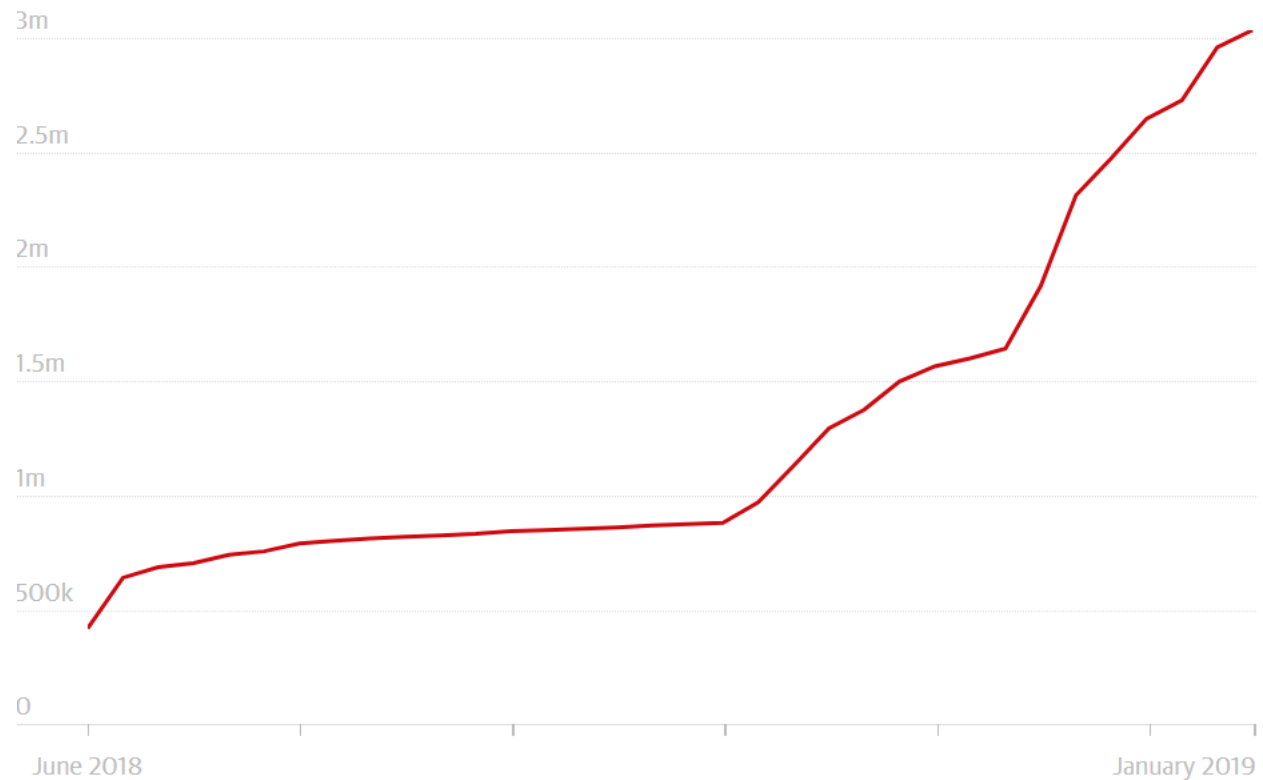
6:00 PM · Oct 13, 2018



[Read the full conversation on Twitter](#)

Alexandra Ocasio-Cortez has seen a 600% increase in Twitter followers since June of last year

Twitter followers



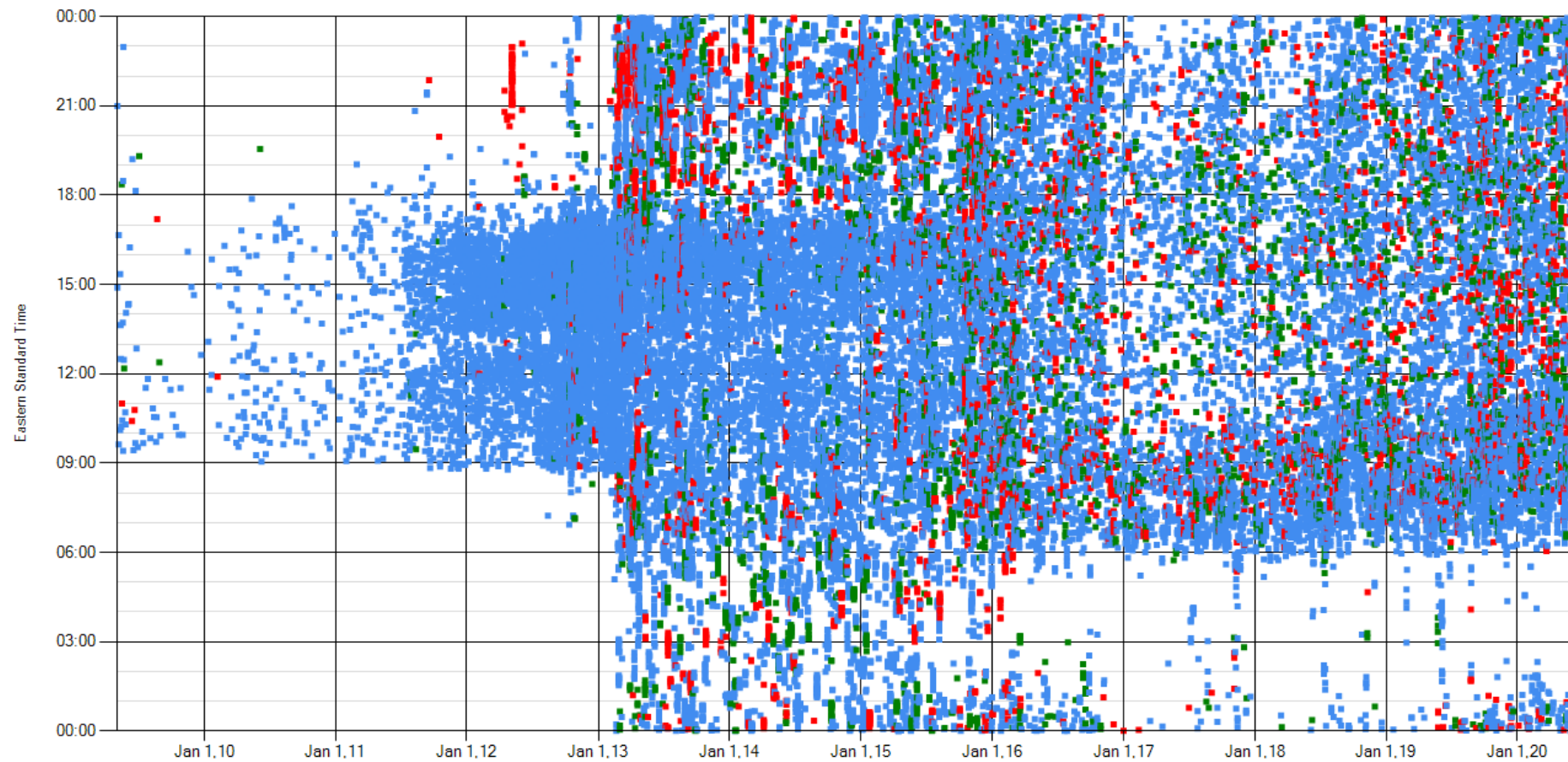
Source: CrowdTangle

Politics also change technology: Twitter (pre-Musk) introduced fact-checking



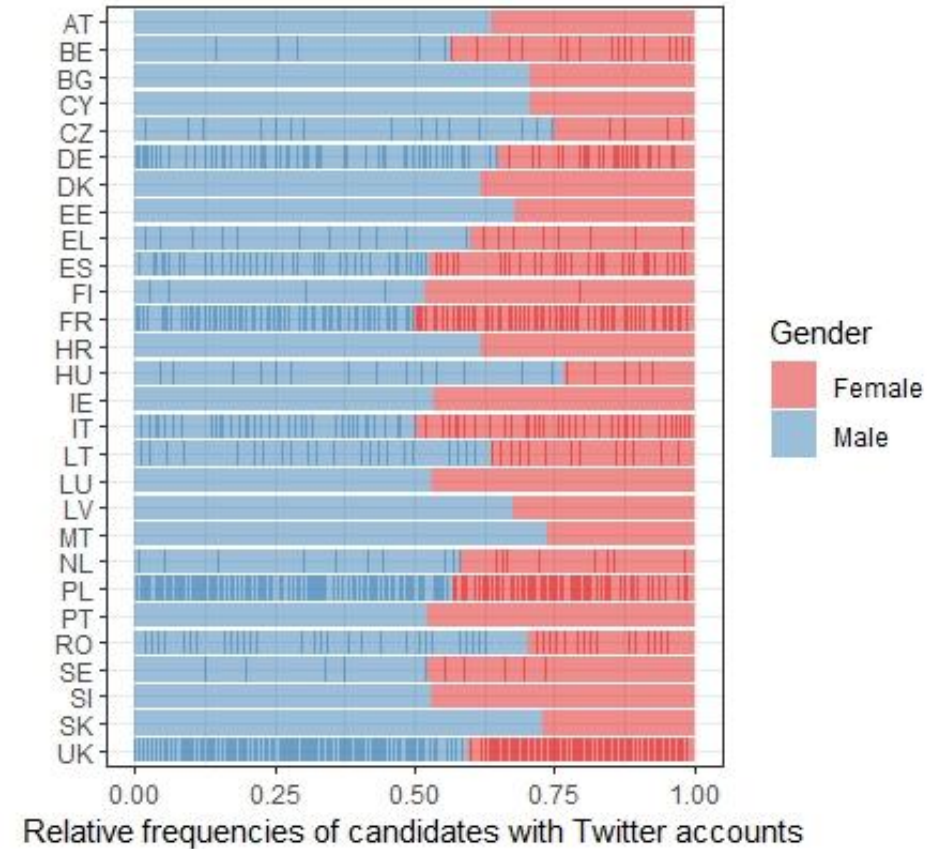
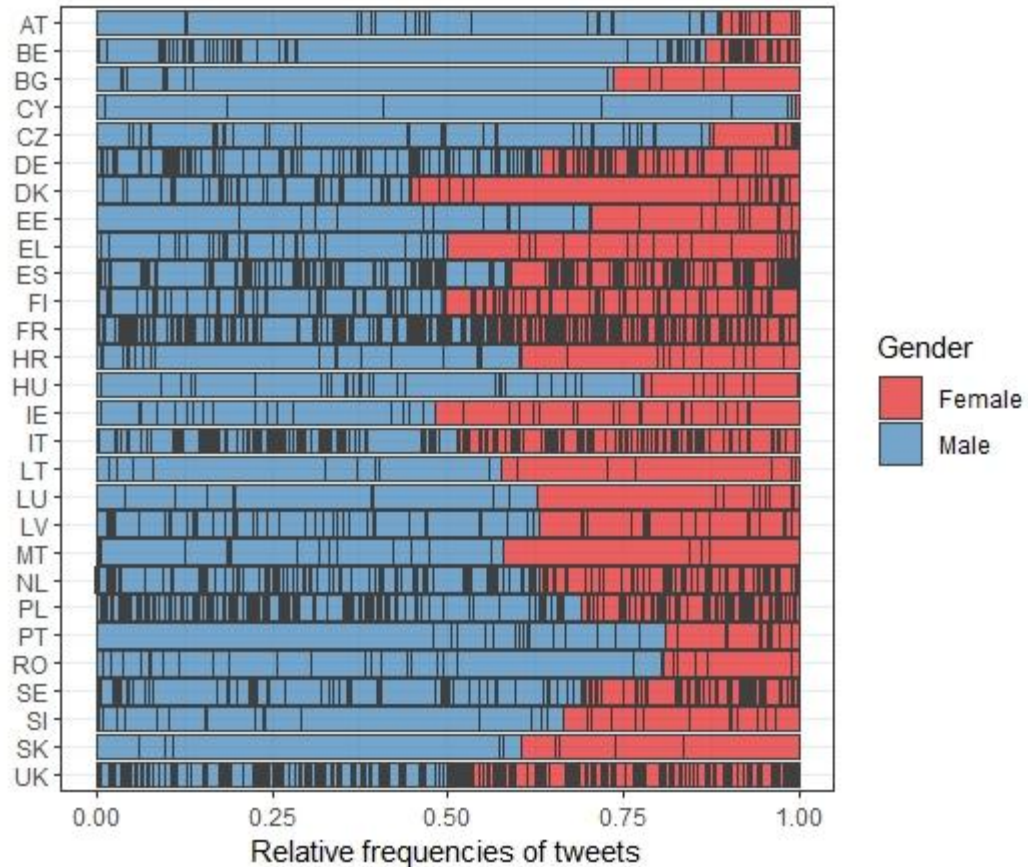
90% of our data is unstructured

Donald J. Trump (@realDonaldTrump) 43436 tweets plotted (11 tweets per day) First tweet=14:54, 4 May 2009
(Red: Sunday Green: Saturday)



Source: https://en.wikipedia.org/wiki/Twitter_use_by_Donald_Trump#/media/File:Twitter_activity_of_Donald_Trump.png

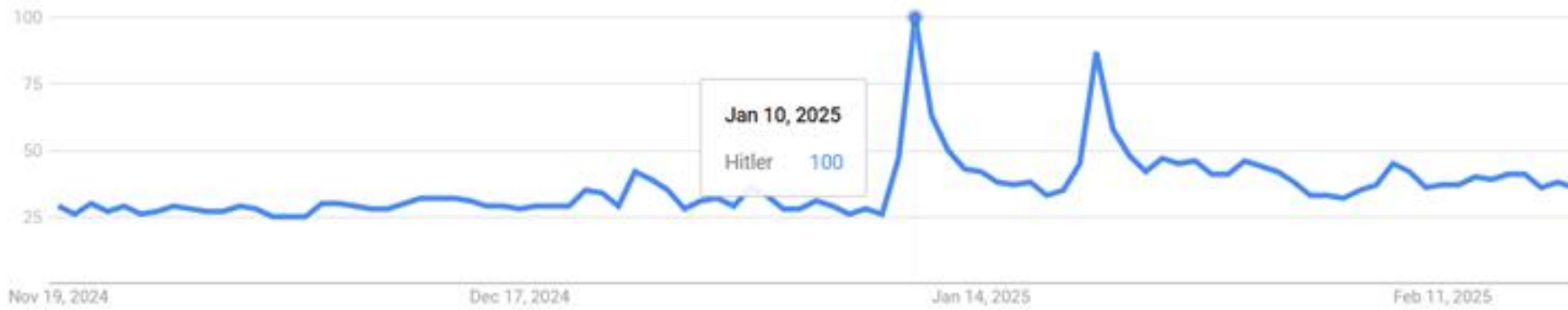
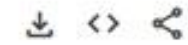
Differences in how male and female candidates campaign?





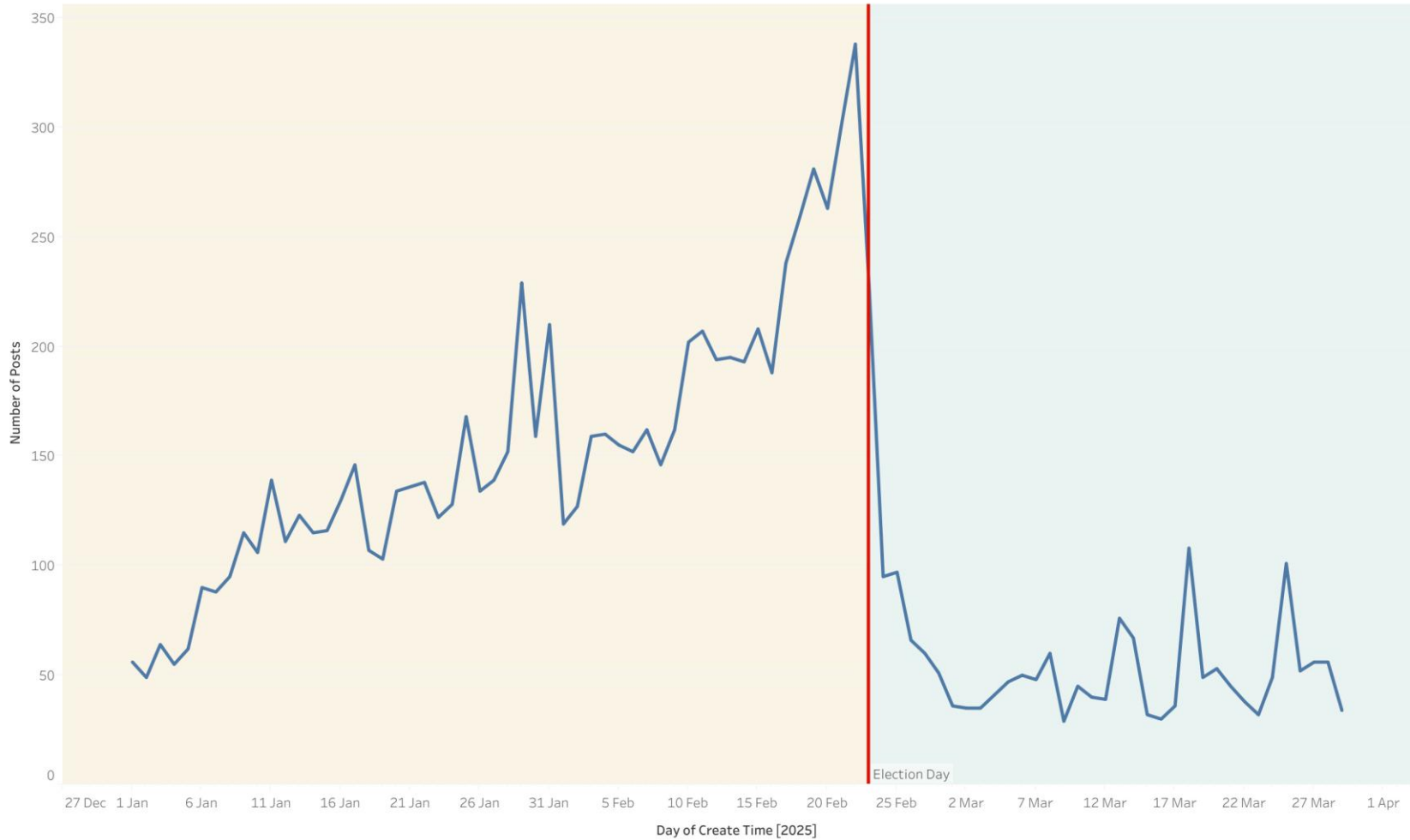
OBSERVATORY OF ONLINE POLITICS

Interest over time ?





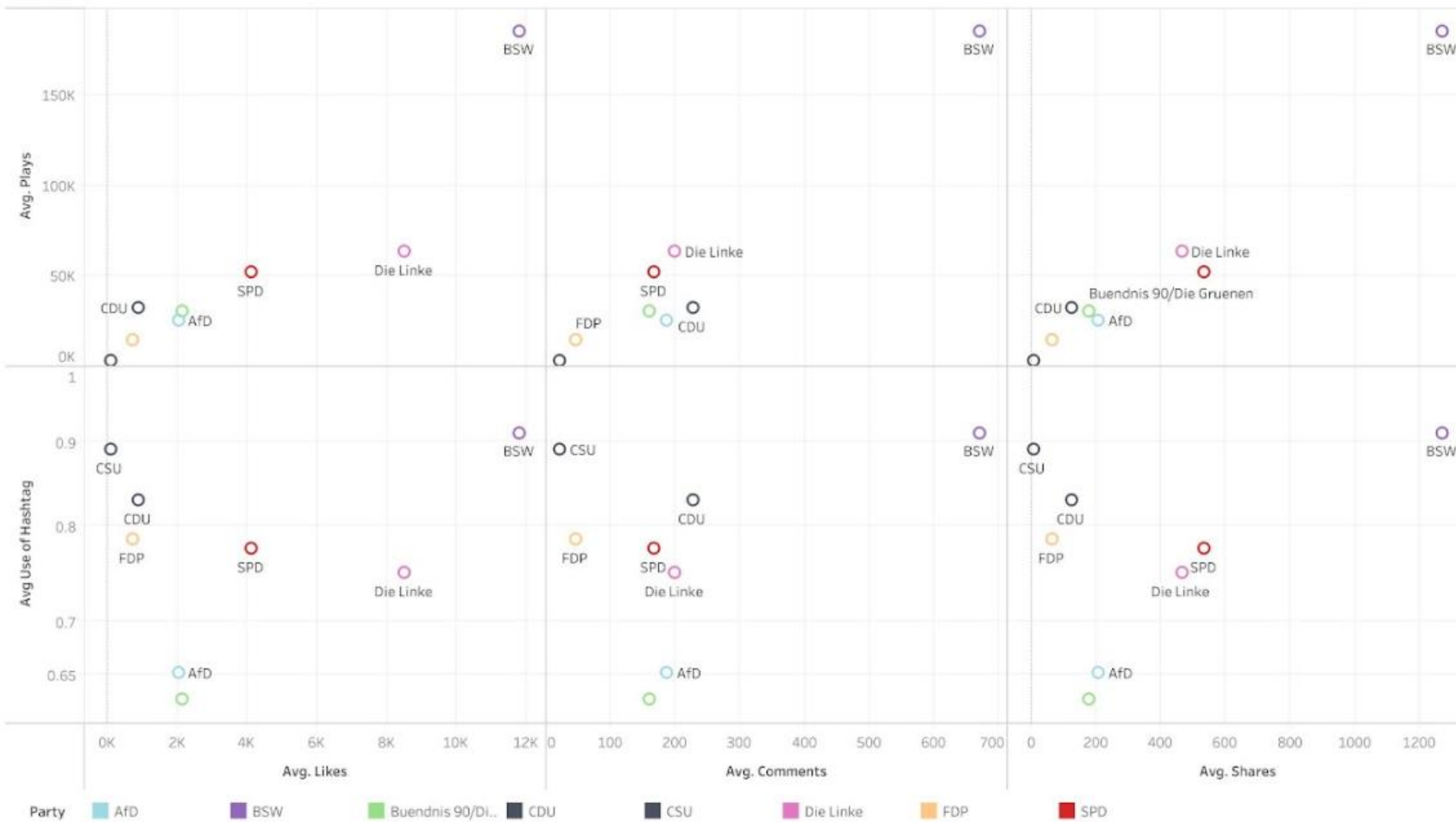
TikTok number of post trend before and after election



Rosa M. Navarrete & Ariq Suryo Hadi P (2025) "TikTok and its role in the 2025 German Elections" in https://politics.i2sc.net/blogposts/tiktok_2025



Engagement Activity



Rosa M. Navarrete & Ariq Suryo Hadi P (2025) "TikTok and its role in the 2025 German Elections" in https://politics.i2sc.net/blogposts/tiktok_2025

We collected the names and the Twitter profiles of the parliamentarians in both upper and lower houses or their equivalent, the ministers of the cabinet, and the political parties in the parliament/government as of April 2023.

Most of the Twitter profiles were obtained from official government. In cases where this was not possible, we looked for Twitter lists made by the Twitter accounts of the Parliament Houses

Of 704 potential accounts, we collected 553 accounts. Of these 553, we consider in our analyses 406 accounts in the period 01.01.2020 – 14.06.2023

The number of tweets analysed is 1,011,593

Measurement

To measure framing we split each tweet in sentences, and classify each sentence using three sentence-level transformer classifiers.

The ManifestoBERTA classifier, covering the 56 categories of the MARPOR coding scheme (multiclass, context-wise).

A finetuned version of the same model, based on environment-related quasi-sentences from the 2019 Euromanifesto project (binary, context-wise).

doc_id
text864385.
text864307.
text863937.
text863340.
text863186.
text864421.
text864260.
text864259.
text864259.
text864244.

Measurement

To measure framing we split each tweet in sentences, and classify each sentence using three sentence-level transformer classifiers

For each sentence we obtain:

- the probability to fall into one of the 56 MARPOR categories
- the probability that the sentence is environment-related

doc_id	marpor_35	marpor_38	marpor_37	marpor_45	env_1
text864385.	0.03	0.94	0.00	0.00	0.89
text864307.	0.94	0.00	0.00	0.00	0.91
text863937.	0.05	0.00	0.00	0.00	0.81
text863340.	0.99	0.00	0.00	0.00	0.94
text863186.	0.95	0.00	0.00	0.00	0.96
text864421.	0.00	0.00	0.88	0.01	0.02
text864260.	0.00	0.00	0.02	0.87	0.02
text864259.	0.00	0.00	0.02	0.94	0.02
text864259.	0.00	0.00	0.03	0.94	0.02
text864244.	0.00	0.00	0.81	0.00	0.01

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the probability that the sentence is environment-related.

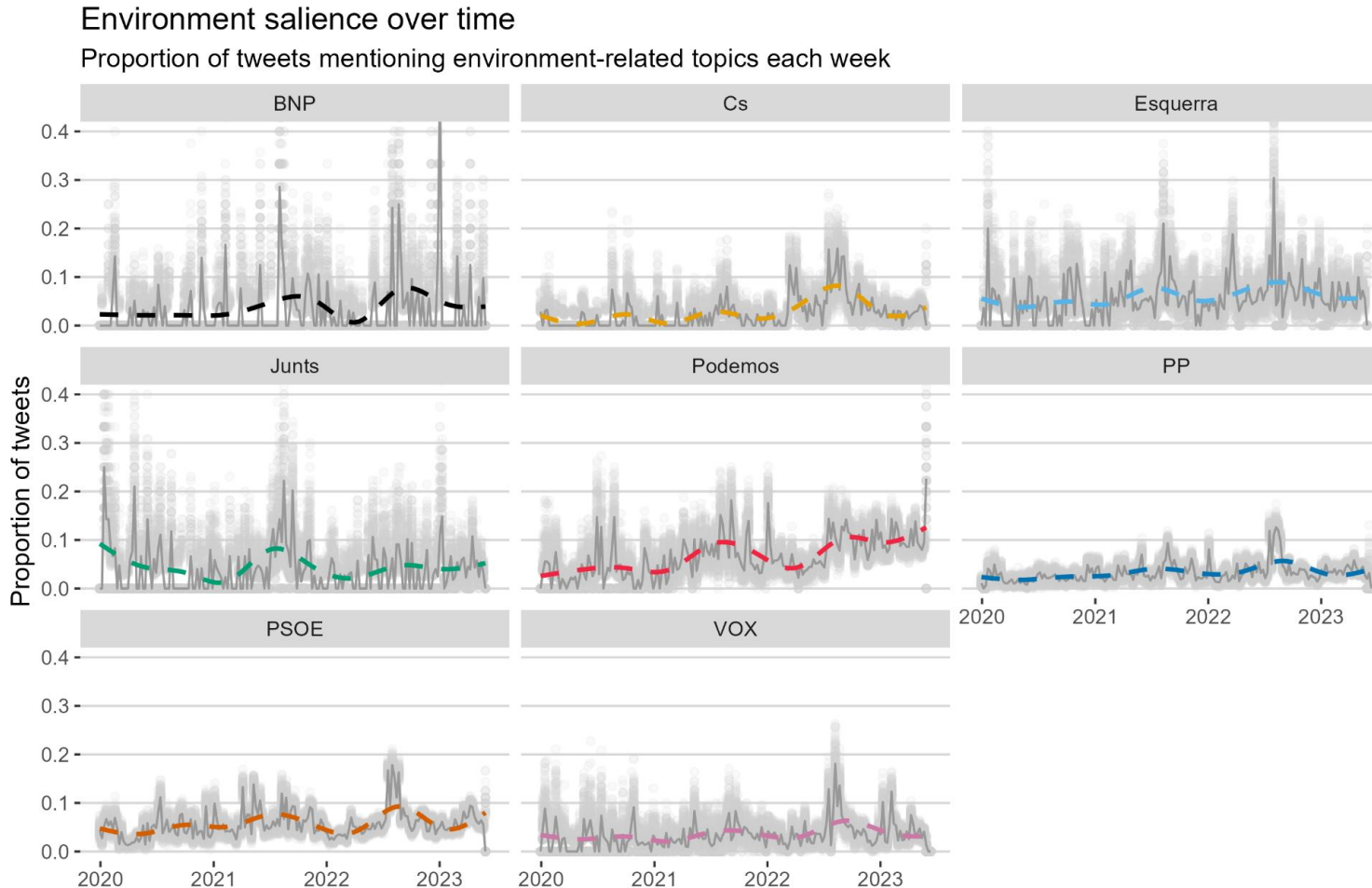
Finally, we apply the following thresholds to obtain the
categories associated with each tweet:

MARPOR .8

Environment .8

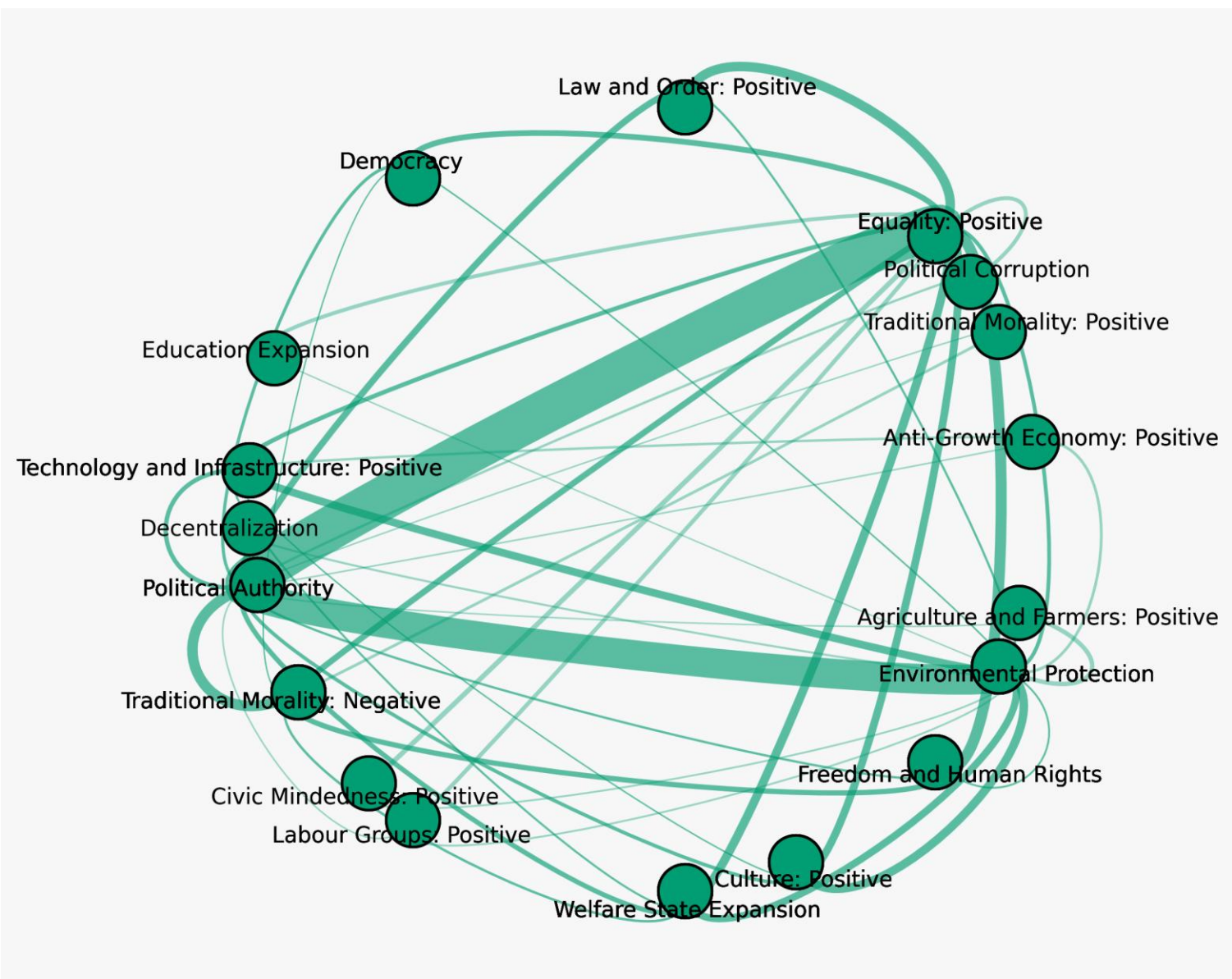
marpor_toj	env_topic
38	1
35	1
34	1
35	1
35	1
37	0
45	0
45	0
45	0
37	0

Results: salience

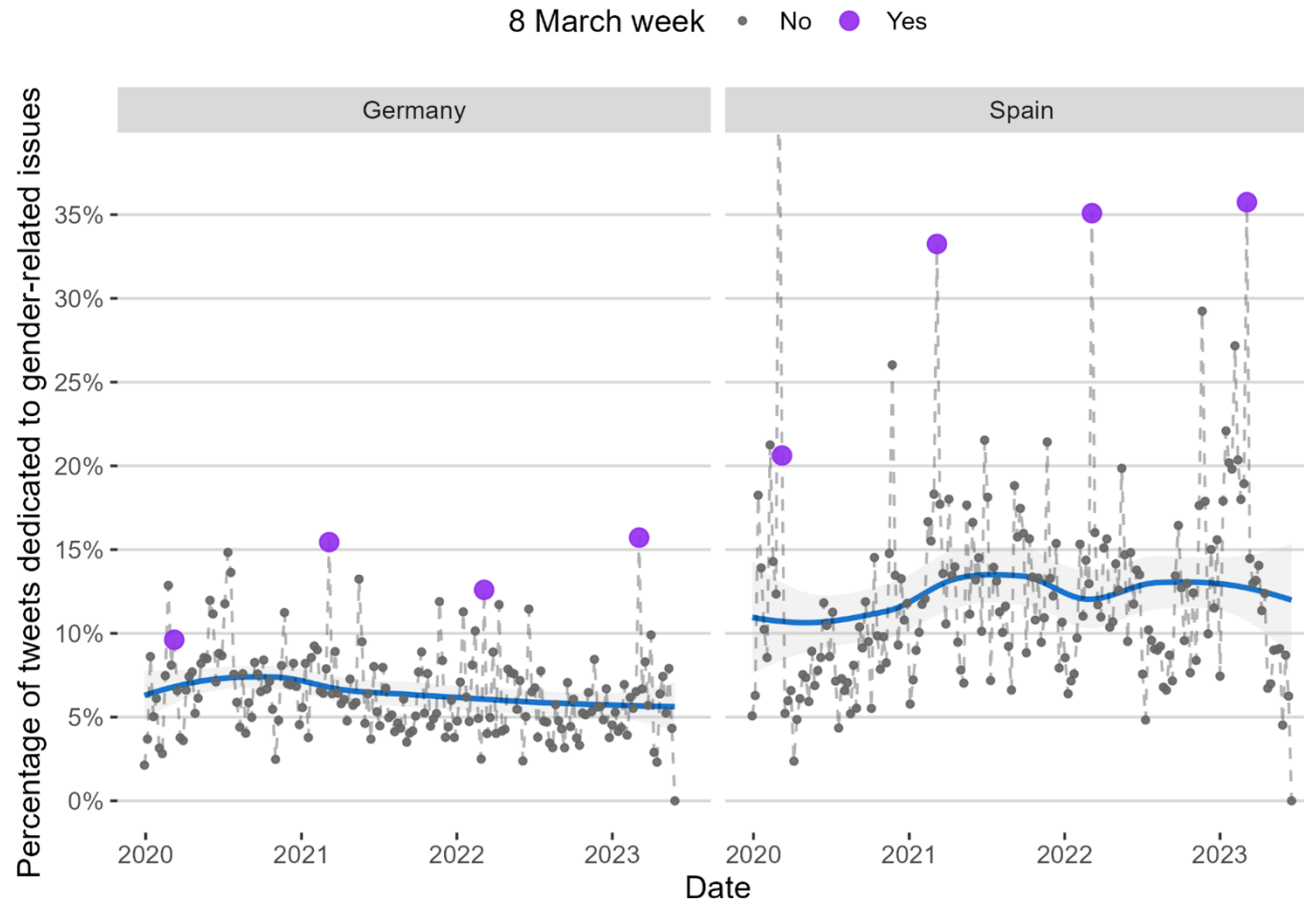


Navarrete, Rosa M., Giuseppe Carteny and Brahmani Nutakki (forthcoming) Framing Environment: A computational analysis of online political communication in Spain

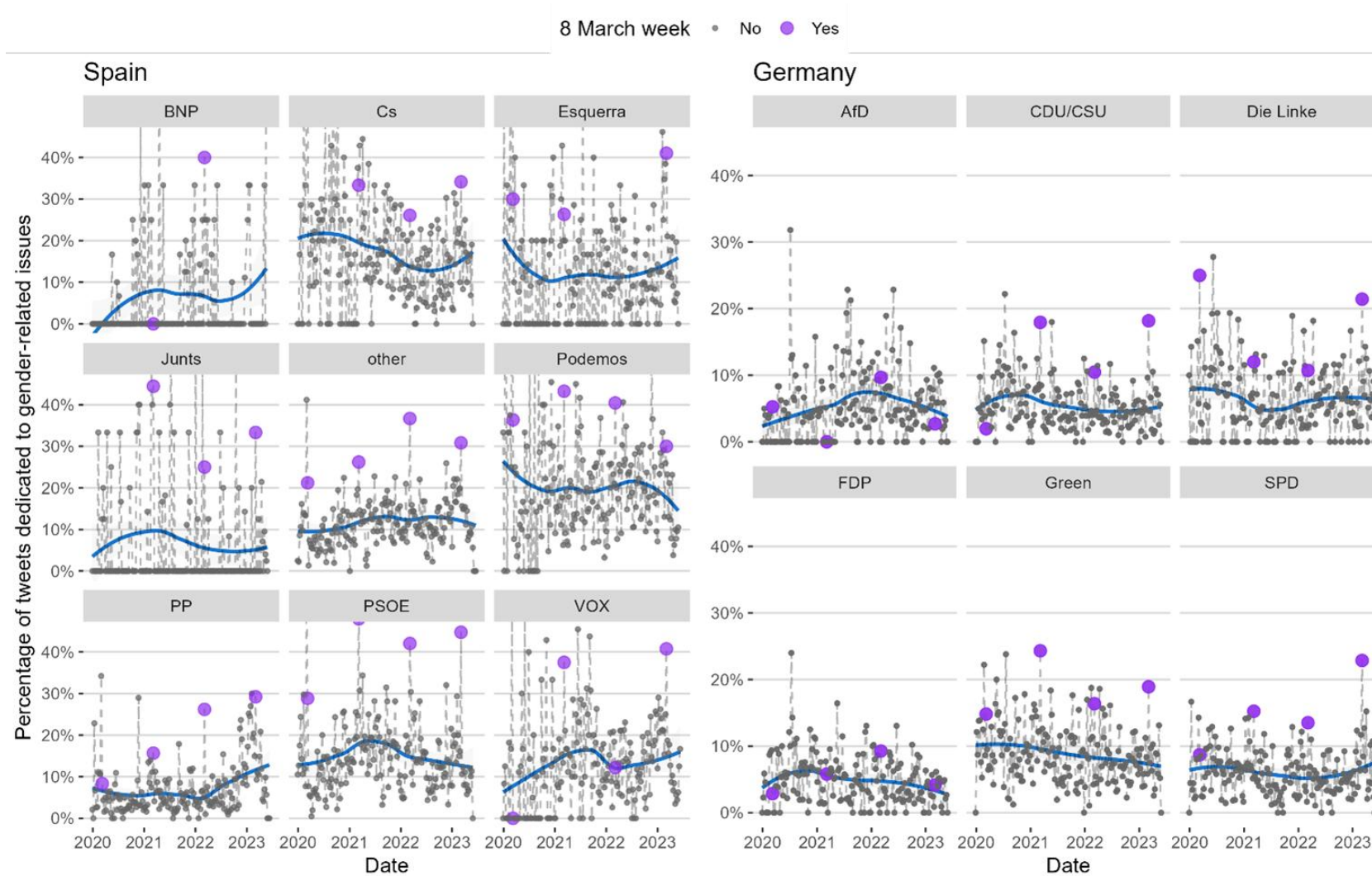
Results: framing



What does the 8th of March tell us about gender and political communication?



Carteny, G. (2025) "What does the 8th of March tell us about gender and political communication?" in <https://politics.i2sc.net/blogposts/8march>



Carteny, G. (2025) "What does the 8th of March tell us about gender and political communication?" in <https://politics.i2sc.net/blogposts/8march>

How It Can Be Done: more innovative ways

The UNTWIST project



Funded by
the European Union

- **LLMs instead of manual coding to improve manifesto research**
- **Still important:** studying political parties' behavior and party competition
 - **Innovation:** using LLMs no (or only little) manual coding is needed
 - **Advantage:** less time for data collection, more time for dealing with important research questions

How It Can Be Done: more innovative ways

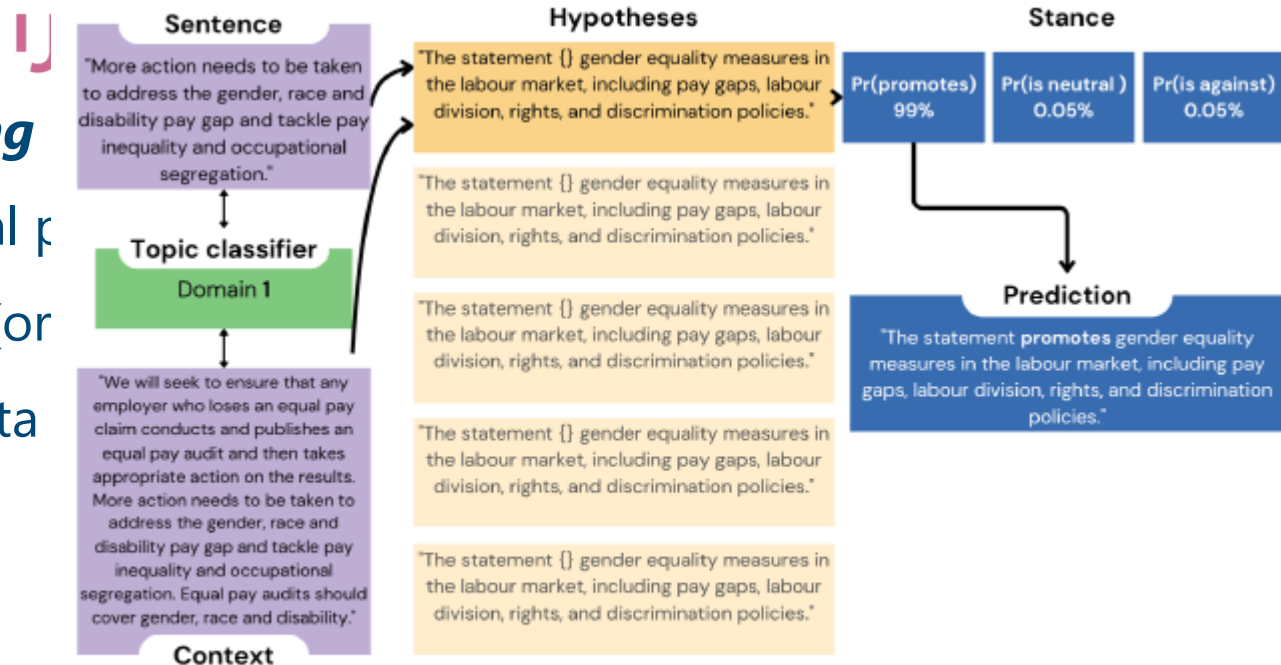
The UNTWIST project



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the European Union

- **LLMs instead of manual coding**
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 - **Innovation:** using LLMs no (or
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Learn more about this in the
afternoon by Giuseppe Carteny



Notes: Example from a British manifesto for the 2015 UK general election.

Figure 3. NLI workflow

How It Can Be Done: more innovative ways

The ActEU project **ActEU** Activating European Citizens' Trust in Times of Crises and Polarisation **Funded by the European Union**

- ***measuring polarization via web and social media data instead of public opinion surveys***
 - *Social media*: Facebook, Twitter/X, WhatsApp-channels
 - *Web*: Media articles and data sources from political actors found online
- ***Advantages and challenges***
 - Polarization is difficult to measure via questions asked in traditional questionnaires
 - Large quantities of text need to be converted into data

How It Can Be Done: more innovative ways

The ActEU project **ActEU** Activating European Citizens' Trust in Times of Crises and Polarisation



Funded by
the European Union

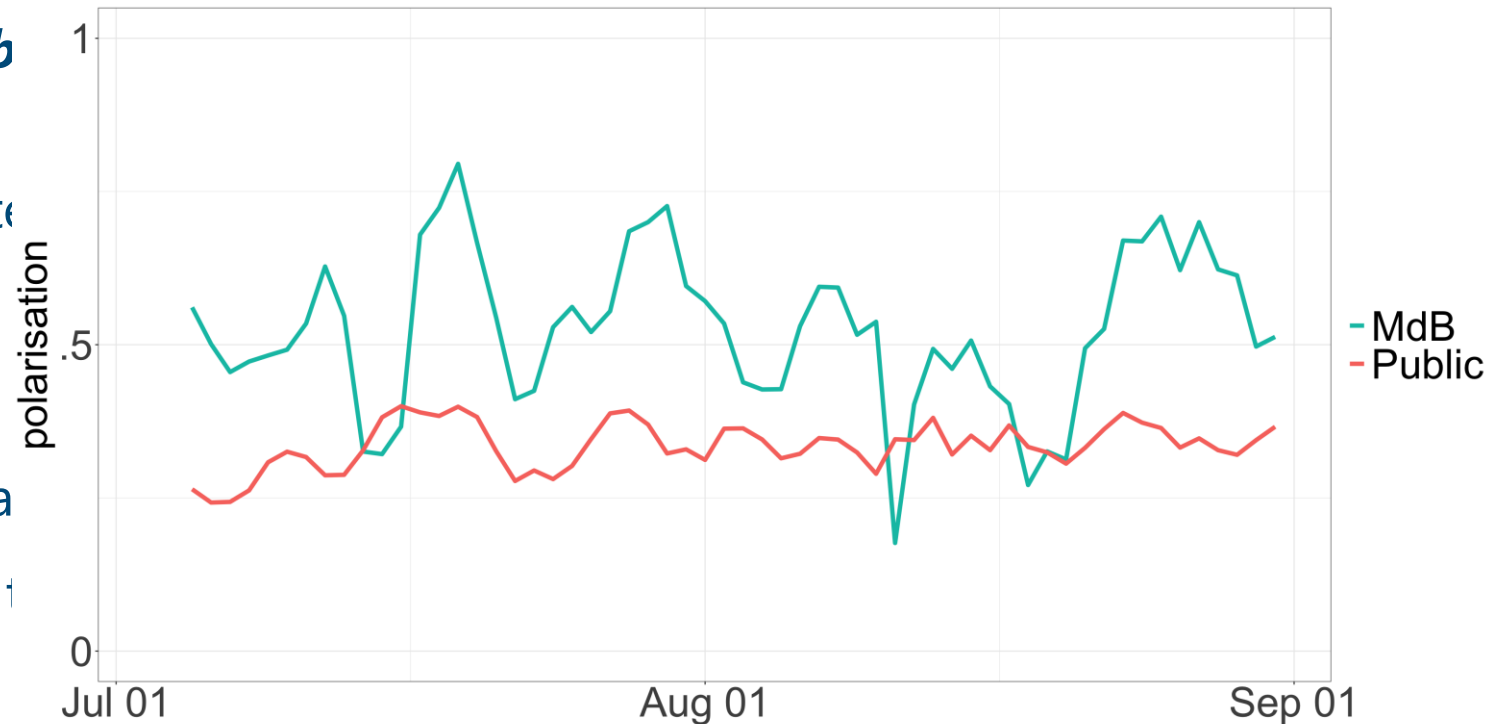
- **measuring polarization via web surveys**

- *Social media*: Facebook, Twitter
- *Web*: Media articles and data

- **Advantages and challenges**

- Polarization is difficult to mea

Learn more about this in the
afternoon by Alex Hartland





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Thanks for your attention.